

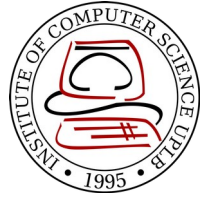
CMSC 137

Data Communications and Networking

Lecture Module
23 – Domain Name System



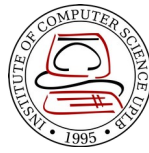
University of the Philippines
LOS BAÑOS



Module 23: Domain Name System

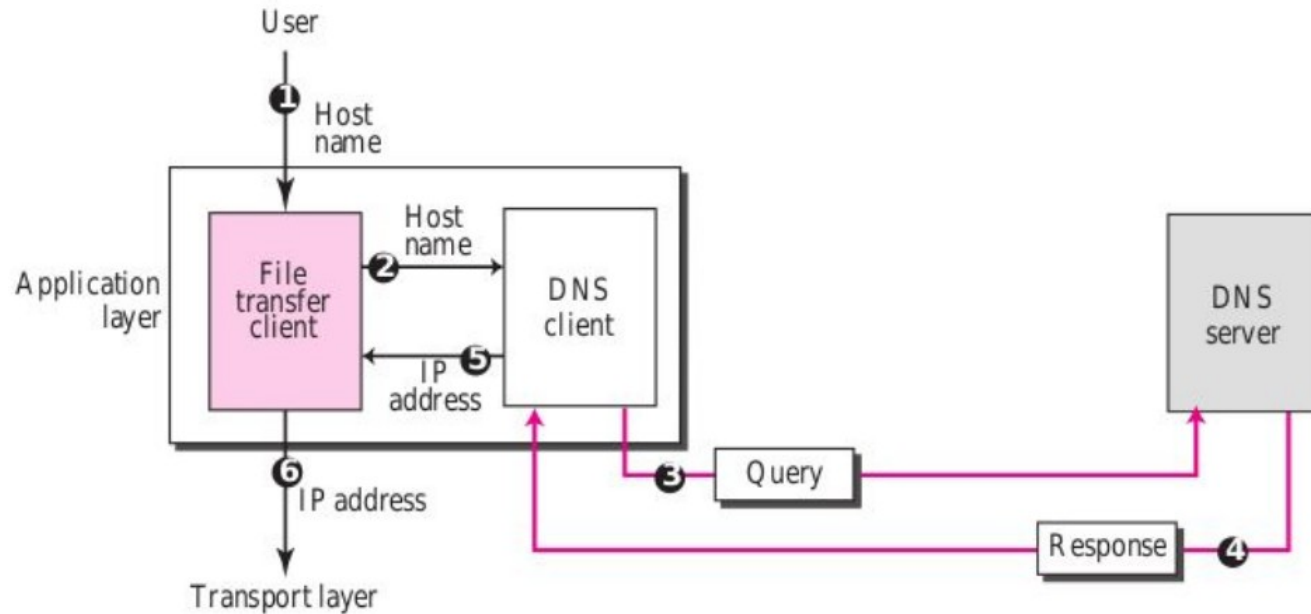
Joseph Anthony C. Hermocilla
Lecturer

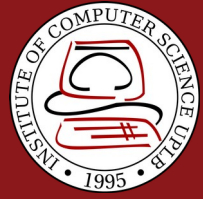
The need for Domain Name System (DNS)



- Names are easier to remember than IP addresses
- A system to map a **hostname** to an IP address or an IP address to a hostname is needed
- Approaches
 - Approach 1: Use a simple host file with two columns: hostname and IP address
 - Approach 2: Dedicate a server to 'serve' the large host file (centralized)
 - Approach 3: "Split" the host file to be served by multiple servers (distributed)

Example: Access a file transfer server given a hostname



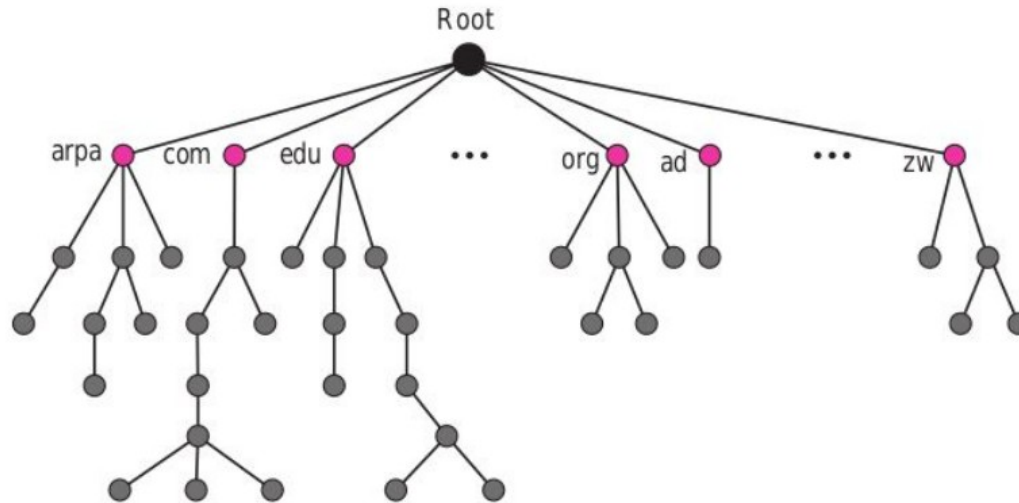


Domain Namespace

Domain Namespace



- **Flat** – names have no structure
- **Hierarchical** - names have several parts, ex. ics.uplb.edu.ph

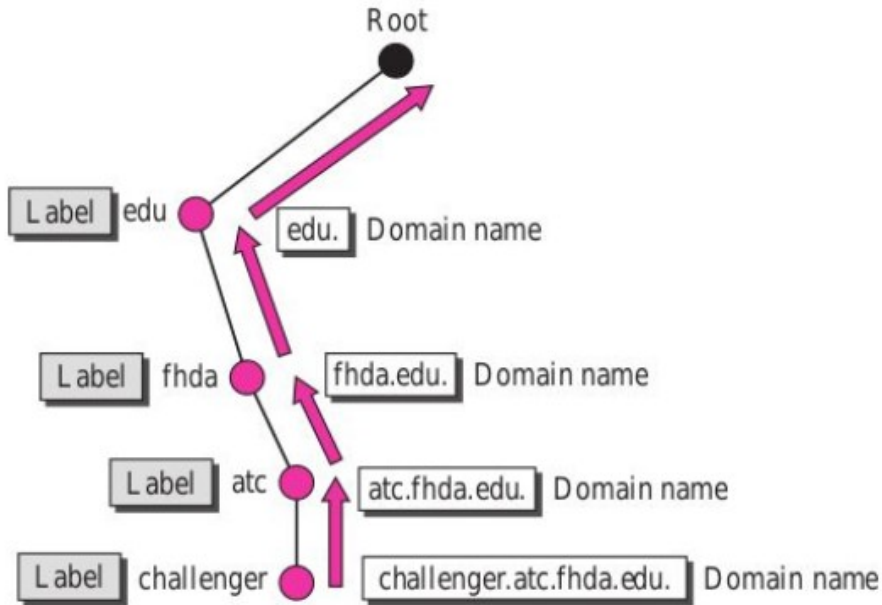


Domain Namespace



- Tree can have only 128 levels: 0 (root) to 127
- **Label** - assigned to a node in the tree, max of 63 characters
 - label for root is the empty string: ""
 - children of a node should have different labels
- **Domain Name (DN)** - a sequence of labels separated by dots (.) up to the root that is assigned to each node
 - **Fully Qualified DN** - last character is a dot
 - **Partially Qualified DN** - last character is not a dot

Domain Namespace



FQDN

challenger.atc.fhda.edu.
cs.hmme.com.
www.funny.int.

PQDN

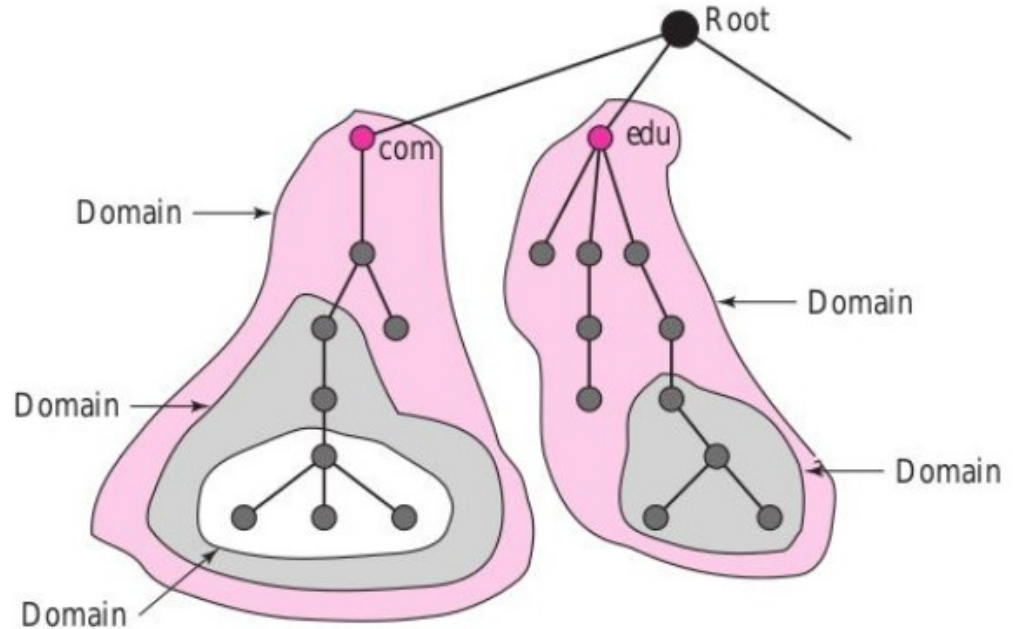
challenger.atc.fhda.edu
cs.hmme
www

Domain Namespace



- Domain

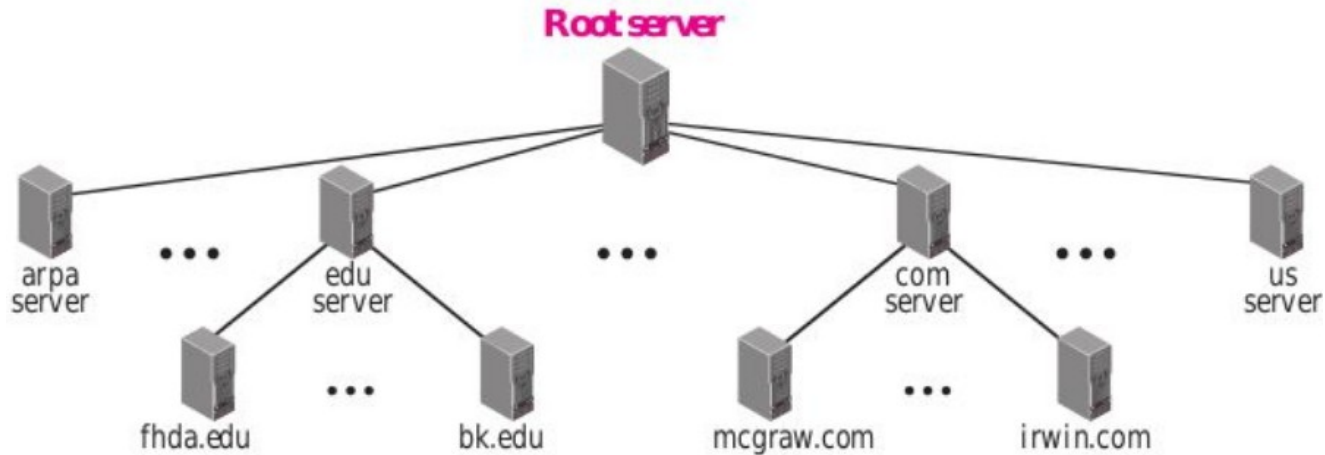
- subtree of the domain namespace
- name of the domain is the domain name of the node at the top of the subtree
- domains may be further divided into subdomains



Distribution of Domain Namespace



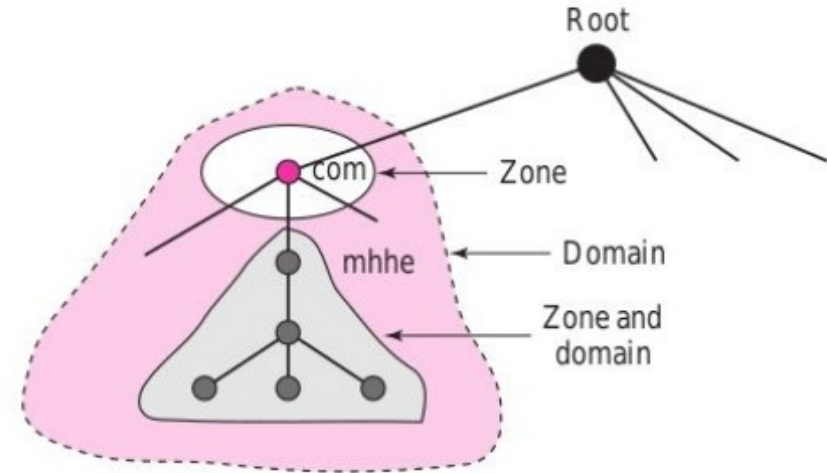
- Information contained in the domain namespace are distributed among many computers called **DNS servers**, organized hierarchically



Distribution of Domain Namespace



- **Zone** - a portion of the domain name space that a DNS server has an authority over or is responsible for
- Domain and Zone can refer to the same thing if a server does not divide the domain into subdomains
- DNS Server maintains a **Zone File** which keeps all information for every node under that domain



Distribution of Domain Namespace



- **Root Server** - a server whose zone consist of the whole domain namespace tree

```
→ dig NS .
; <> DiG 9.18.18-0ubuntu0.22.04.2-Ubuntu <> NS .
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 42247
;; flags: qr rd ra; QUERY: 1, ANSWER: 13, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;      IN      NS

;; ANSWER SECTION:
.      87203      IN      NS      j.root-servers.net.
.      87203      IN      NS      h.root-servers.net.
.      87203      IN      NS      d.root-servers.net.
.      87203      IN      NS      m.root-servers.net.
.      87203      IN      NS      l.root-servers.net.
.      87203      IN      NS      f.root-servers.net.
.      87203      IN      NS      g.root-servers.net.
.      87203      IN      NS      c.root-servers.net.
.      87203      IN      NS      b.root-servers.net.
.      87203      IN      NS      i.root-servers.net.
```

```
→ dig NS ph @e.root-servers.net
; <> DiG 9.18.18-0ubuntu0.22.04.2-Ubuntu <> NS ph @e.root-servers.net
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 13487
;; flags: qr rd; QUERY: 1, ANSWER: 0, AUTHORITY: 4, ADDITIONAL: 9
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1472
;; QUESTION SECTION:
;ph.      IN      NS

;; AUTHORITY SECTION:
ph.      172800    IN      NS      1.ns.ph.
ph.      172800    IN      NS      ph.communitydns.net.
ph.      172800    IN      NS      ns2.cuhk.edu.hk.
ph.      172800    IN      NS      ns4.apnic.net.

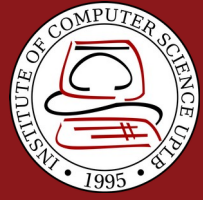
;; ADDITIONAL SECTION:
1.ns.ph. 172800    IN      A      206.51.255.1
1.ns.ph. 172800    IN      AAAA   2620:171:805:ad2:7068::1
ph.communitydns.net. 172800    IN      A      194.0.1.23
ph.communitydns.net. 172800    IN      AAAA   2001:678:4::17
ns2.cuhk.edu.hk. 172800    IN      A      137.189.6.21
```

Distribution of Domain Namespace



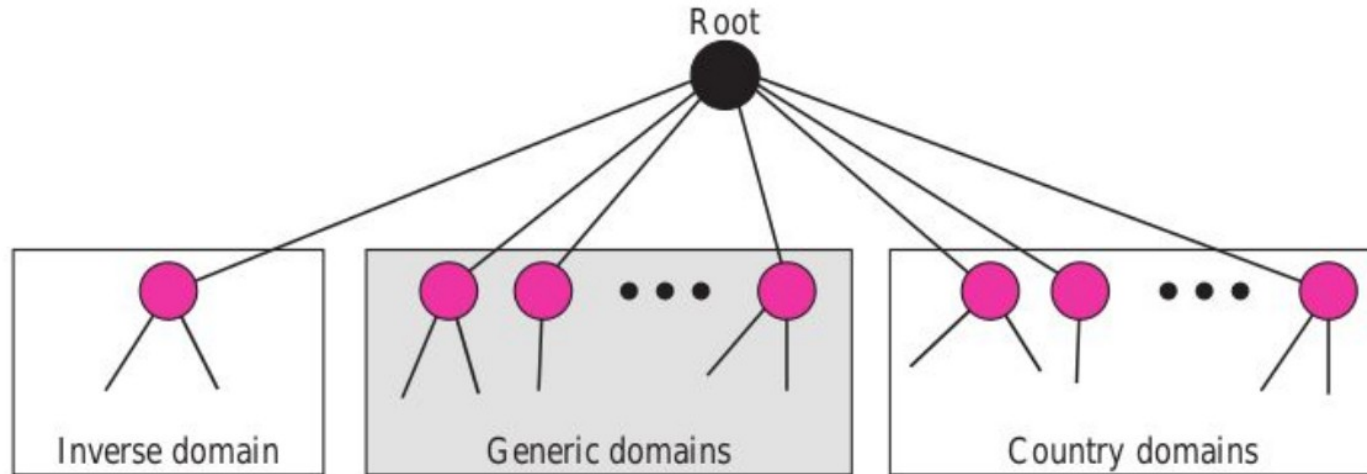
- **Primary DNS server** - responsible for creating, maintaining, and updating the zone file for which it is an authority
- **Secondary DNS server** - transfers the complete information about a zone from another server (**Zone Transfer**); does not update the zone file; for redundancy

```
tmp
→ dig NS jachermocilla.org +short
ns1.digitalocean.com.
ns3.digitalocean.com.
ns2.digitalocean.com.
tmp
→ dig NS blog.jachermocilla.org +short
ghs.google.com.
tmp
→
```



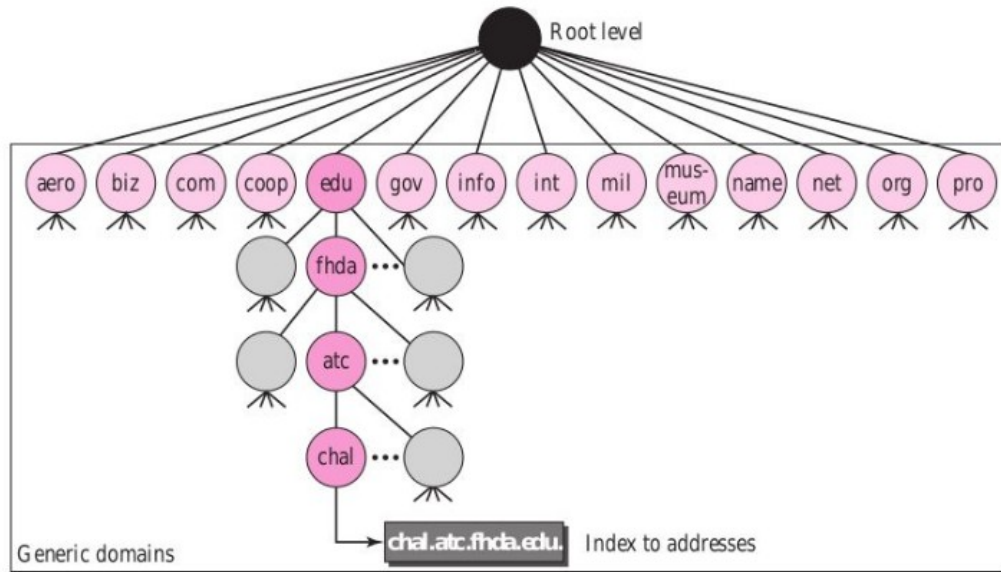
DNS in the Internet

Domains in the Internet



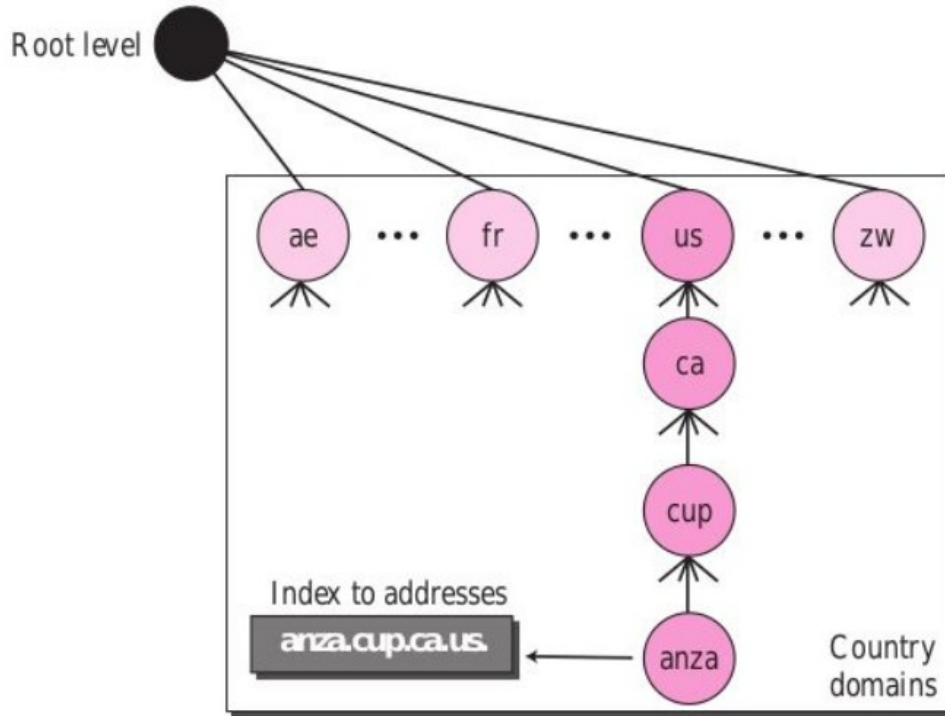
<https://data.iana.org/TLD/tlds-alpha-by-domain.txt>

Generic Domains

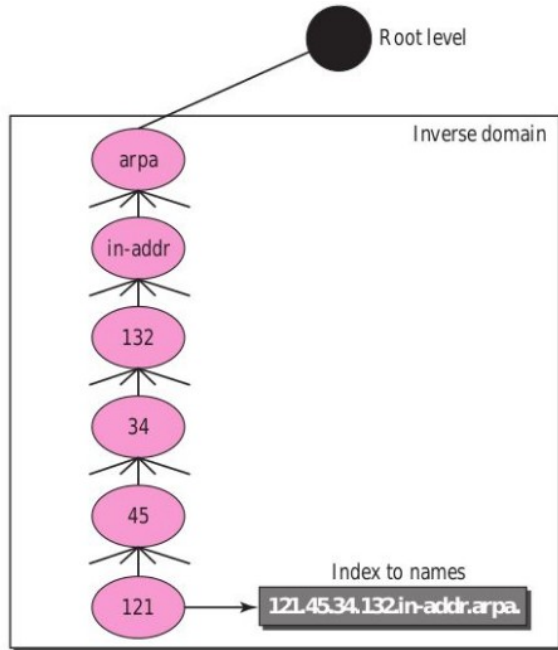


Label	Description
aero	Airlines and aerospace companies
biz	Businesses or firms (similar to "com")
com	Commercial organizations
coop	Cooperative business organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International organizations
mil	Military groups
museum	Museums and other non-profit organizations
name	Personal names (individuals)
net	Network support centers
org	Nonprofit organizations
pro	Professional individual organizations

Country Domains



Inverse Domains



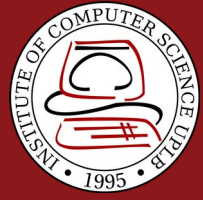
```
tmp
→ dig -x 157.230.45.190

; <> DiG 9.18.18-0ubuntu0.22.04.2-Ubuntu <> -x 157.230.45.190
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 54519
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;190.45.230.157.in-addr.arpa. IN PTR

;; ANSWER SECTION:
190.45.230.157.in-addr.arpa. 1706 IN PTR jachermocilla.org.

;; Query time: 0 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Tue Apr 16 16:35:23 PST 2024
;; MSG SIZE rcvd: 87
```



Name Resolution

DNS Resolver



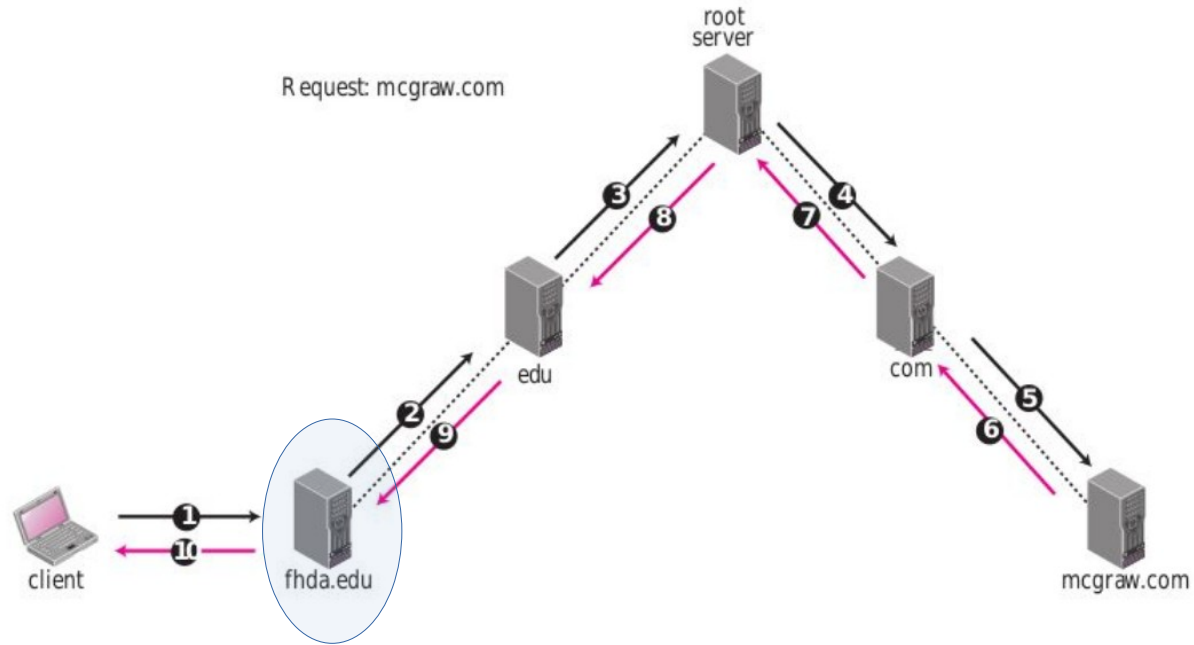
- Also called a **DNS client**
- checks the cache or contacts a DNS server
 - in linux DNS server information is placed traditionally in `/etc/resolv.conf`
 - In modern Linux distros, it is managed by Systemd

```
# Run "resolvectl status" to see details about the uplink DNS servers
# currently in use.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

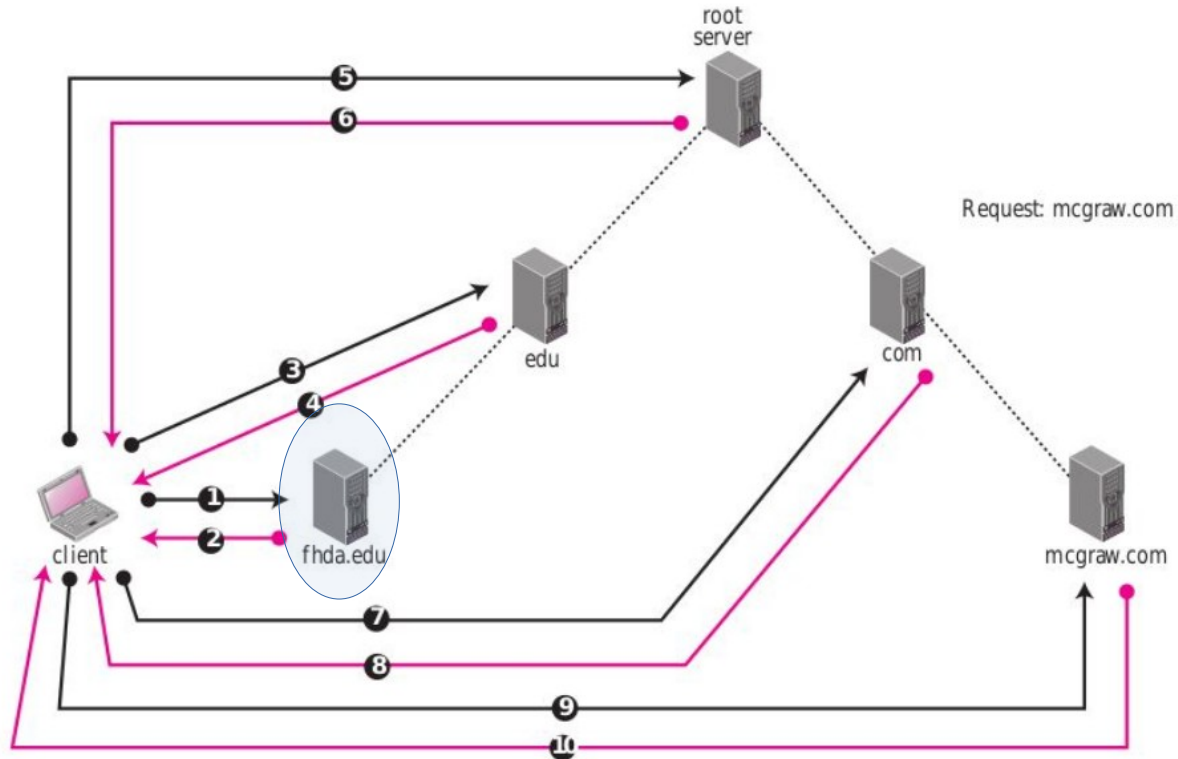
nameserver 127.0.0.53
options edns0 trust-ad
search .
tmp
→ resolvectl status
Global
    Protocols: -LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
resolv.conf mode: stub

Link 2 (wlp2s0)
    Current Scopes: DNS
    Protocols: +DefaultRoute +LLMNR -mDNS -DNSOverTLS DNSSEC=no/unsupported
    Current DNS Server: 192.168.1.1
    DNS Servers: 192.168.1.1
```

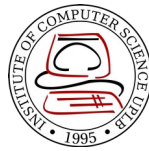
Recursive Resolution



Iterative Resolution



dig +trace

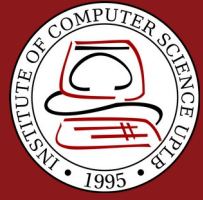


```
→ dig jachermocilla.org +trace

; <=> DiG 9.18.18-0ubuntu0.22.04.2-Ubuntu <=> jachermocilla.org +trace
;; global options: +cmd
.          3664      IN      NS      m.root-servers.net.
.          3664      IN      NS      f.root-servers.net.
.          3664      IN      NS      j.root-servers.net.
.          3664      IN      NS      b.root-servers.net.
.          3664      IN      NS      g.root-servers.net.
.          3664      IN      NS      e.root-servers.net.
.          3664      IN      NS      h.root-servers.net.
.          3664      IN      NS      i.root-servers.net.
.          3664      IN      NS      l.root-servers.net.
.          3664      IN      NS      d.root-servers.net.
.          3664      IN      NS      a.root-servers.net.
.          3664      IN      NS      c.root-servers.net.
.          3664      IN      NS      k.root-servers.net.
;; Received 239 bytes from 127.0.0.53#53(127.0.0.53) in 0 ms

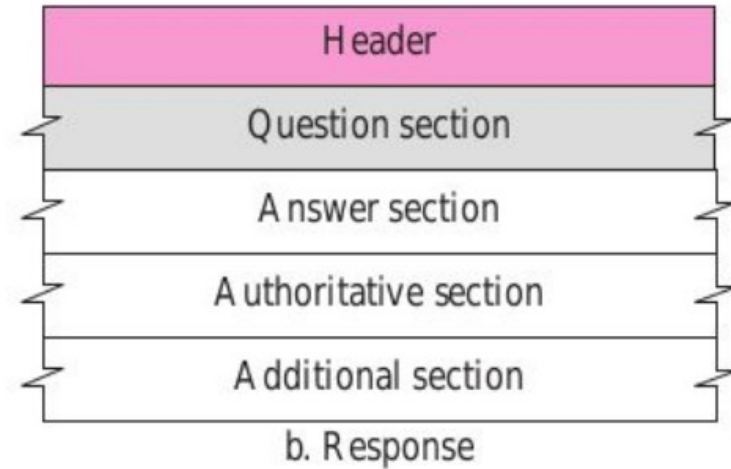
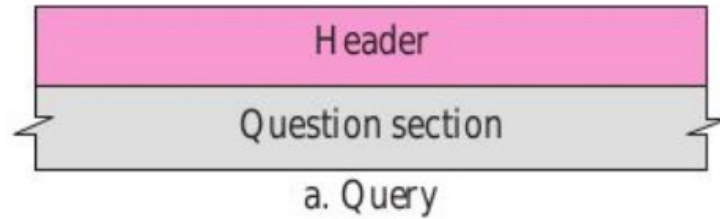
org.        172800   IN      NS      b2.org.afiliast-nst.org.
org.        172800   IN      NS      b0.org.afiliast-nst.org.
org.        172800   IN      NS      d0.org.afiliast-nst.org.
org.        172800   IN      NS      a0.org.afiliast-nst.info.
org.        172800   IN      NS      a2.org.afiliast-nst.info.
org.        172800   IN      NS      c0.org.afiliast-nst.info.
org.        86400   IN      DS      26974 8 2 4FEDE294C53F438A158C41D39
1DE32
org.        86400   IN      RRSIG   DS 8 1 86400 20240429050000 2024041
GRLDXtC41wVKfmHaw+jFXnHvFh0diFYxI+M2 de35QjrV5ah3mv1jNpQ/ZL11Lv07Bz05Mwmgk5gyZL+xzt
ya0FWCLw12tHp61jNzq7uDVbWA/PIkNL Zwuz4305oMxQ4Hgo0vpHYyh1VFXGQHSE6R+qL0DxN5J4DBgP9/
opeXIKU/vYzq05aD+V4kqa/nr+Eh b71Uh1BL5DJKhfmHmu5mhl8ogy2pHEe2JWJPBVkGEyDhhRNaaJBVwy
;; Received 786 bytes from 202.12.27.33#53(m.root-servers.net) in 27 ms

jachermocilla.org. 3600      IN      NS      ns2.digitalocean.com.
```



DNS Messages

Query and Response Messages



Header



- Size is 12 bytes, same for query and response
- **Identification** - 16-bit field use by client to match response with query

Identification	Flags
Number of question records	Number of answer records (All 0s in query message)
Number of authoritative records (All 0s in query message)	Number of additional records (All 0s in query message)

Header

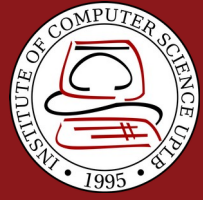


- Flags – 16 bits



- QR (query/response) - 0 if query, 1 if response
- OpCode - 4 bits, defines the type of query or response (0-standard, 1-inverse, 2-status request)
- AA (authoritative answer) - 1 bit, 1 if server is authoritative
- TC (truncated) - 1 bit, 1 if response was more than 512 bytes
- RD (recursion desired) - 1 bit, 1 if recursive answer is desired
- RA (recursion available) - 1 bit, 1 if recursive is available
- Reserved - three 0's
- rCode - 4 bits

Value	Meaning	Value	Meaning
0	No error	4	Query type not supported
1	Format error	5	Administratively prohibited
2	Problem at name server	6–15	Reserved
3	Domain reference problem		

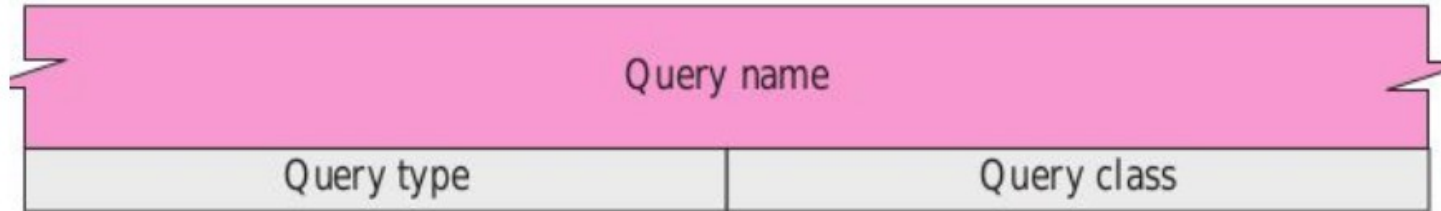


Types of Records

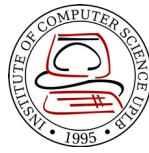
Question Record



- Used by client to get information from server



Question Record



- Query Name

Count						Count				Count				Count				Count	
5	a	d	m	i	n	3	a	t	c	4	f	h	d	a	3	e	d	u	0

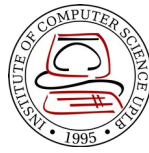
Question Record



- Query Type

Type	Mnemonic	Description
1	A	Address. A 32-bit IPv4 address. It converts a domain name to an address.
2	NS	Name server. It identifies the authoritative servers for a zone.
5	CNAME	Canonical name. It defines an alias for the official name of a host.
6	SOA	Start of authority. It marks the beginning of a zone.
11	WKS	Well-known services. It defines the network services that a host provides.
12	PTR	Pointer. It is used to convert an IP address to a domain name.
13	HINFO	Host information. It defines the hardware and operating system.
15	MX	Mail exchange. It redirects mail to a mail server.
28	AAAA	Address. An IPv6 address (see Chapter 26).
252	AXFR	A request for the transfer of the entire zone.
255	ANY	A request for all records.

Question Record



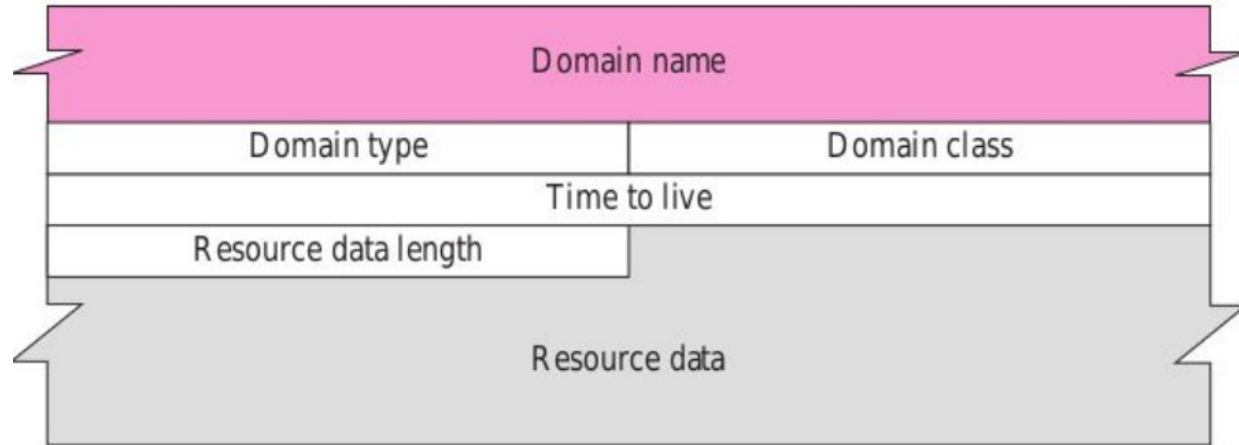
- Query Class

<i>Class</i>	<i>Mnemonic</i>	<i>Description</i>
1	IN	Internet
2	CSNET	CSNET network (obsolete)
3	CS	The COAS network
4	HS	The Hesiod server developed by MIT

Resource Record



- Associated with each domain name is a **Resource Record**



Resource Record



- **Domain name** - same as from question record
- **Domain type** - same as query type field in question record
- **Domain class** - same as query class field in the question record
- **Time-to-live** - 32-bit field that the number of seconds the answer is valid, 0 means not cached
- **Resource data length** - 16 bits to define the length of resource data
- **Resource data** - variable length containing the answer to the query: number, domain name, offset pointer, character string

References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.